

Civics Group	Index Number	Name (use BLOCK LETTERS)
--------------	--------------	--------------------------



ST. ANDREW'S JUNIOR COLLEGE
JC2 PRELIMINARY EXAM
2008

H1 BIOLOGY

8875/2

Paper 2: Structured and Free Response

11 September 2008

Thursday

2 hours

Additional Materials: Answer Paper
 Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
 Write in dark blue or black pen.
 You may use a soft pencil for any diagram, graph or rough working.
 Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination,

1. Hand up Section A and Section B **separately**.
2. **Write the question number** attempted for Section B on the Cover Sheet.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/12
2	/ 8
3	/10
4	/10
Section B	/20
Total	/60

QUESTION 1

The Figure 1.1 below shows a model of the enzyme chymotrypsin. Chymotrypsin catalyses the hydrolysis of peptide bonds of proteins. The active site of chymotrypsin is located in a slight depression on one side of the molecule. Three amino acids form the active site. These three amino acids are some distance apart on the polyptide chain but close together in the active site.

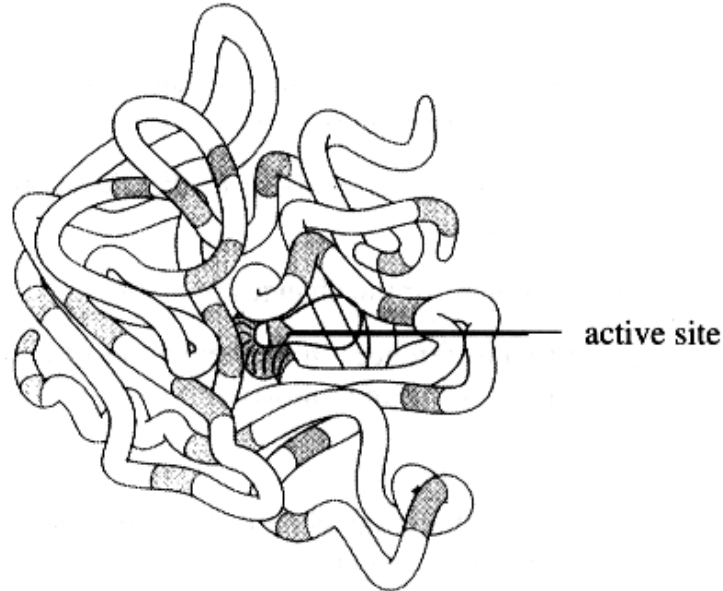


Figure 1.1

- (a) Describe how amino acid residues at different positions in the protein may be brought together in the active site when the enzyme is synthesised.

.....

.....

.....

.....[2]

- (b) Explain how a substrate may be attached to the enzyme.

.....

.....

.....

.....

.....

.....[3]

- (d)** Complete the table below to show 3 differences between inhibitors in Graph 2 and Graph 3 and their mode of action. [3]

Graph 2	Graph 3

[Q1 Total: 12]

QUESTION 2

Silena alba (White Champion), there are separate male and female plants. Male plants are XY and female plants are XX.

In 1912, Baur described the first example of sex-linkage in plants. He discovered an unusual form of *Silena alba* with narrow leaves. It was found to be a male plant. Pollen from this plant was used to fertilise a female plant with normal broad leaves, and all progeny, male and female, had normal broad leaves. When these normal broad-leaved plants were cross fertilised, they produce 167 plants with broad leaves and 60 with narrow leaves.

When the narrow-leaved plants were grown to maturity all were male, while of the broad-leaved plants, 56 were male and 111 were female.

(a) Draw a genetic diagram to explain both crosses. [5]

(b) Explain why there were no female narrow-leaved plants.

.....

[2]

(c) Suggest how a female narrow-leaved plant could be produced.

.....
[1]

[Q2 Total: 8]

QUESTION 3

The polymerase chain reaction is a process used to make many copies of a piece of DNA containing the insulin gene. The process is outline in the figure 3.1.

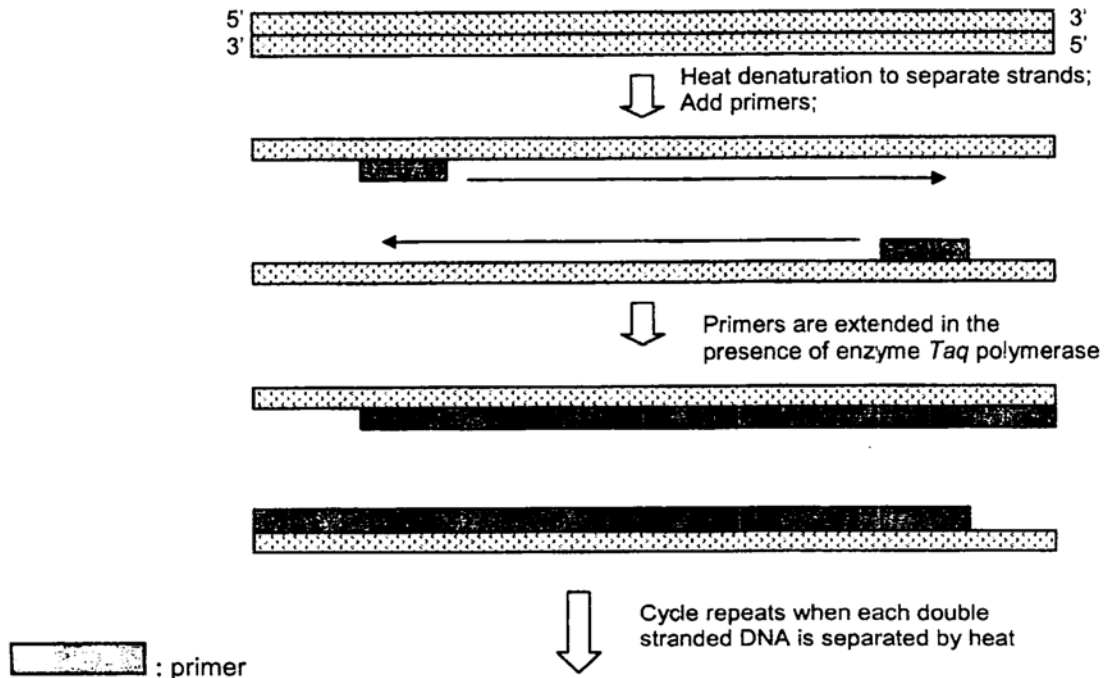


Figure 3.1

- (a) The primer is a short sequence of nucleotides that binds to the DNA. Explain why the primer only binds to the DNA at a particular position.

.....
[1]

- (b) Describe the role of the primer in the PCR.

.....
[1]

- (c) Both processes, polymerase chain reaction and DNA semi-conservative replication involves elongation of polynucleotide chains. List 2 differences between the polymerase chain reaction and DNA semi-conservative replication.

.....

[2]

The cloned DNA and plasmids were first cut by the same restriction enzyme and then mixed together with DNA ligase. Several types of plasmids were formed. Some contained the human DNA in the centre of the gene coding for resistance to tetracycline. The different types of plasmids are shown in Figure 3.2.

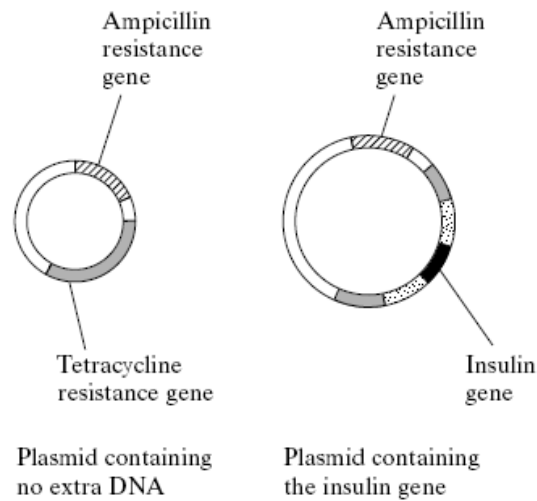


Figure 3.2

- (e) What are the properties of restriction enzymes that make them useful in genetic engineering?

.....

[2]

- (d) The plasmids are mixed with bacteria. Some bacteria took up the plasmids. Explain how it is possible to distinguish between bacteria which have taken up a plasmid with the insulin gene and those which have taken up a plasmid without any insulin gene.

.....

[4]

[Q3 Total: 10]

QUESTION 4

The Figure 4.1 below shows the flow of electrons in the non-cyclic and cyclic photophosphorylation.

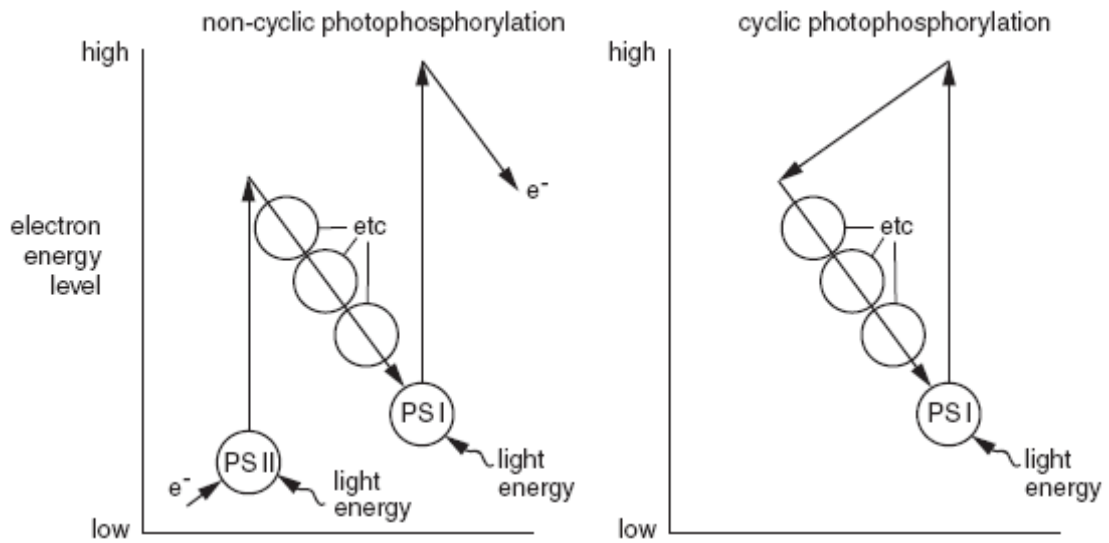


Figure 4.1

(a) State the precise location of:

- (i) Photophosphorylation - [1/2]
- (ii) oxidation of NADPH - [1/2]

(b) Describe how non-cyclic photophosphorylation differs from cyclic photophosphorylation.

Non-cyclic photophosphorylation	Cyclic photophosphorylation

(c) Briefly describe the main stages of the dark reaction.

.....

.....

.....

.....

.....

.....

.....

.....[4]

(d) Explain briefly what initially happens to the concentration of RuBP and GP if the supply of carbon dioxide is reduced.

.....

.....

.....

.....[2]

[Q4 Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 5** (a) Describe the unique features of zygotic stem cells, embryonic stem cells and blood stem cells. [12]
- (b) Describe how mitosis maintains genetic stability and its importance in growth, repair and asexual reproduction. [8]

[Total: 20]

- 6** (a) Describe the structure and roles of ribosome and transfer RNA in the synthesis of proteins. [10]
- (b) Using a named example, explain how natural selection may bring about evolution. [10]

[Total: 20]